

Growing Peppers in the Dragoon Mountains

A Complete Hydroponic Pepper Growing Guide

*Chiltepin · Jalapeño · Poblano · Cayenne · Serrano · Rocoto
Habanero · Ancho · Hatch · Bell & Sweet Varieties
DWC · Dutch Bucket · NFT · Masterblend · pH · EC · Harvest*

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Why the Dragons Are Perfect for Peppers

Native Chiltepin · climate heritage · your natural advantages

The Dagoon Mountains are not just suitable for pepper growing — they are ancestral pepper territory. The Chiltepin (*Capsicum annuum* var. *glabriusculum*), the wild ancestor of nearly all cultivated chiles, grows natively in Cochise County and the surrounding sky island ranges. The Apache people harvested and used it for centuries. You are growing peppers in their homeland.

Climate Advantages for Pepper Growing

Factor	Dagoon Conditions	Why Peppers Love It
Summer temperatures	85–95°F daytime	Ideal fruiting range. Hot enough to set fruit reliably without blossom drop from extreme heat.
Night temperatures	55–70°F — cool nights	Peppers set fruit best when nights drop below 75°F. Dagoon nights are perfect.
Monsoon humidity	50–70% July–Sept	Boosts fruit sizing and prevents blossom drop from dry air stress.
Monsoon rainfall	8–12" in 6 weeks	Reduces irrigation need by 40–60% during peak fruiting season.
Diurnal temperature swing	30–40°F daily	Concentrates capsaicin and flavour compounds. Dagoon chiles are hotter and more flavourful.
Fall season	Sept–Nov — warm days, cool nights	Extended ripening window. Peppers continue producing long after summer crops are done.
UV intensity	15–20% higher at elevation	Accelerates ripening and colour development. Intense sun deepens flavour.
Frost timing	First frost Oct–Nov at mid elevation	Long growing season — 180+ frost-free days at 4,500–5,000 ft.

The Chiltepin Connection

The wild Chiltepin grows along rocky canyon washes and under mesquite trees throughout Cochise County, including the Dagoon and Chiricahua mountain ranges. These small, intensely hot berries (50,000–100,000 SHU) disperse via birds and have survived here for thousands of years with zero irrigation and no amendments. They prove this land was made for peppers.

★ **Local tip:** Wild Chiltepin plants can be found growing naturally along canyon washes in the Dragons. Collect seeds from ripe berries in fall — these are perfectly adapted to your exact conditions and make excellent parent stock for your garden. They are also genuinely rare and delicious.

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Pepper Species & Variety Guide

Scoville ratings · days to maturity · Dragoon suitability

Peppers belong to five main domesticated species. Understanding which species each variety belongs to helps predict its temperature preferences, growing habit, and how well it will perform at Dragoon elevation.

Species Overview

Species	Examples	Temp Preference	Dragoon Suitability
C. annuum	Jalapeño, poblano, cayenne, serrano, bell, Hatch, Anaheim, Chiltepin	Hot summers, warm nights	★★★★★ Excellent — widest variety selection, most productive here
C. chinense	Habanero, Scotch bonnet, ghost pepper	Very hot — needs high heat	★★★★ Good — lower Dragoon elevations best; may need shade cloth above 5,500 ft
C. pubescens	Rocoto, Manzano	Cool highlands — prefers 45–65°F nights	★★★★★ Ideal — the Dragoons' cool nights are exactly what Rocoto needs
C. frutescens	Tabasco, Malagueta	Hot, humid	★★★ Moderate — monsoon season works well; dry season challenging
C. baccatum	Aji Amarillo, Lemon Drop	Warm, moderate	★★★★ Good — Andean origin tolerates elevation and temperature swings

Complete Variety Reference Guide

Variety	Species	SHU	Heat	Days	Height	Dragoon Notes
Bell Pepper	C. annuum	0	Sweet	70–80	18–24"	Excellent — fall crop. Heat above 90°F causes blossom drop; plant for fall harvest.
Poblano / Ancho	C. annuum	1,000–2,000	Mild	65–80	24–30"	★ Excellent. Thick walls love the Dragoon fall. Green = poblano; dried red = ancho.
Anaheim / Hatch	C. annuum	500–2,500	Mild	75–80	24–30"	★ Excellent. New Mexico Hatch varieties thrive at similar elevations. Roast by the bushel.
Hungarian Wax	C. annuum	1,000–15,000	Mild–Medium	70	18–24"	Good. Early ripening — useful for spring harvest before monsoon.
Jalapeño	C. annuum	2,500–8,000	Medium	70–75	24–30"	★ Excellent. Native to Mexico — loves the Dragoon climate. Hotter when stressed. Very productive.

Serrano	C. annuum	10,000–25,000	Hot	75–90	18–24"	★ Excellent. More heat-tolerant than jalapeño. Prolific producer. Great for fresh salsa.
Chipotle (smoked jalapeño)	C. annuum	2,500–8,000	Medium	70–75	24–30"	Harvest jalapeños when red for traditional chipotle smoking over mesquite — perfect local wood.
Cayenne	C. annuum	30,000–50,000	Hot	70–75	18–24"	★ Excellent. Heat-tolerant, prolific, and easy to dry at Dragoon elevation. Thin walls dry perfectly in the fall sun.
Chiltepin (native)	C. annuum	50,000–100,000	Very Hot	90–120	36–48"	★★ Native to Cochise County — zero adaptation needed. Perennial if protected from frost. Grows wild in canyon washes. The most important local variety.
Thai / Bird's Eye	C. annuum	50,000–100,000	Very Hot	75–90	18–24"	Good. Compact plant, prolific, heat-tolerant. Excellent container variety.
Habanero	C. chinense	100,000–350,000	Extreme	85–100	24–36"	Good at lower elevations (< 5,000 ft). Needs long, hot season. Start 10–12 weeks early. Monsoon humidity improves fruit set.
Ghost / Bhut Jolokia	C. chinense	800,000–1,000,000+	Extreme	100–120	36–48"	Challenging — needs very long season. Start January indoors. Best below 4,800 ft elevation. Monsoon season critical for fruit set.
Scotch Bonnet	C. chinense	100,000–350,000	Extreme	90–100	24–36"	Moderate — similar to habanero. Fruity flavour worth the effort.
Rocoto (Red/Yellow)	C. pubescens	30,000–100,000	Very Hot	90–120	36–60"	★★ Ideal — the Dragoons' cool nights are perfect for Rocoto. Perennial to 15+ years if protected from frost. See dedicated Chapter 3.
Manzano (Apple Chile)	C. pubescens	12,000–30,000	Hot	120+	36–60"	★★ Ideal — same as Rocoto. Yellow/orange apple-shaped fruit. Black seeds. Beautiful purple flowers.
Aji Amarillo	C. baccatum	30,000–50,000	Hot	90–100	36–48"	Good. Andean origin — elevation tolerant. Fruity, unique flavour. Excellent food pepper. Needs staking.
Lemon Drop	C. baccatum	15,000–30,000	Hot	90–100	24–36"	Good. Citrusy, bright flavour. Heat and drought tolerant. Excellent for salsas and hot sauce.

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The Rocoto Special Section

Capsicum pubescens · high-elevation adaptation · growing secrets

The Rocoto (*Capsicum pubescens*) is the rarest, most elevation-adapted, and most misunderstood pepper species in cultivation. Native to the Andes Mountains at 6,000–14,000 ft elevation, it is the only pepper species that thrives in cool conditions — and it is perfectly suited to the Dragoon Mountains. While most chiles struggle with cool Dragoon nights, the Rocoto absolutely requires them to set fruit.

Why Rocoto is Perfect for the Dragons

Rocoto Requirement	Dragoon Condition	Match?
Cool nights 45–65°F for fruit set	55–68°F night temps most of the year	★★ Perfect
Partial shade tolerance in hot weather	Mountain aspect provides afternoon shade on east-facing slopes	★★ Perfect
Well-drained, deep soil	Sandy loam over caliche — break through hardpan at planting	✓ Good
pH 6.0–7.0	Dragoon soils pH 7.5–8.5 — needs sulfur amendment	■ Needs work
Moderate humidity during fruit set	Monsoon July–Sept provides 50–70% humidity	★★ Perfect
Protection from frost (perennial)	Frost arrives Oct–Nov — needs protection or harvest by then	✓ Manageable
Long growing season 90–120+ days	180+ frost-free days at mid elevation	✓ Sufficient

Rocoto Growing Protocol for the Dragons

1	Start very early — January indoors — Rocotos need 10–14 weeks before transplant. Germination is slow (14–28 days). Use bottom heat mat at 75–80°F.
2	Partial shade siting — Plant on the east side of a structure or taller plants. Morning sun is fine — afternoon shade prevents blossom drop in summer heat.
3	Amend soil heavily — Rocoto needs pH 6.0–6.5. Apply sulfur in fall before planting. Add 4" compost. They are heavy feeders compared to other chiles.
4	Consistent moisture — Never let Rocotos dry out — this immediately causes blossom drop. Daily drip irrigation during dry periods. Mulch heavily.
5	Patience at flowering — Rocotos bloom prolifically but will drop flowers if nights exceed 75°F. The Dragoon monsoon season (cool nights returning) triggers the best fruit set.

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Frost protection = perennial — At the first frost threat in October, cover with frost cloth or bring container plants indoors. Protected Rocotos can live 15+ years and grow 10 ft tall — becoming a permanent garden feature.

Black seeds: Rocoto is the only domesticated pepper species with black seeds — a reliable identifier. It also has distinctly hairy leaves and stems, and beautiful purple-violet six-pointed star flowers. If your 'rocoto' has white flowers and white/tan seeds, it is mislabelled — check your seed source.

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Hydroponic System Selection

DWC · Dutch Bucket · NFT — matched to each pepper variety

Peppers are among the most rewarding hydroponic crops — they produce abundantly, respond immediately to precise nutrition, and can be grown year-round in the Dragoons with LED supplementation. The right system depends on the variety's size and root mass. Large fruiting peppers like Rocoto and habanero need Dutch Bucket; smaller chiles like jalapeño and cayenne thrive in DWC.

System Recommendations by Variety

System	Best Pepper Varieties	Container / Setup	Dragoon Notes
Dutch Bucket ★ Best for large chiles	Rocoto, Habanero, Poblano, Bell, Ghost	5-gallon buckets with perlite media. 1 plant per bucket.	Best system for the Dragoons — insulate buckets against temperature swings. Bury partially to keep root zone below 72°F.
DWC (Deep Water Culture)	Jalapeño, Serrano, Cayenne, Chiltepin, Thai	5-gallon dark buckets with 3" net pots and airstone.	Simplest system. Excellent for medium peppers. Critical: keep reservoir below 72°F — insulate or bury.
NFT (Nutrient Film)	Chiltepin, Thai, Serrano, small chiles	4" channels, 12–18" plant spacing.	Best water efficiency — important in the Dragoons. Not suited for large varieties with heavy root mass.
Kratky (Passive DWC)	Jalapeño, Serrano, Cayenne, herbs	Mason jar or 1-gallon container. No pump needed.	Excellent off-grid option. Works well outdoors in shade during monsoon. Refill every 5–7 days with fresh nutrient solution.
Ebb & Flow	Most medium varieties	Tray with timer and pump. Perlite or hydroton media.	Good for multiple smaller plants. Monitor EC carefully — solution concentrates faster in Dragoon heat.

Reservoir Temperature — Critical in the Dragoons

Root zone temperature is the #1 system management challenge in the high desert. Roots above 72°F lose dissolved oxygen, become susceptible to Pythium root rot, and stop absorbing nutrients efficiently — even when the solution is perfectly balanced.

- **Bury or insulate reservoirs** — underground tanks stay 10–15°F cooler. Even wrapping in foam insulation makes a significant difference.
- **Use dark/opaque containers** — prevents algae and reduces heat absorption
- **Cover all reservoir surfaces** — evaporation in the high desert is extreme; an open 5-gallon bucket loses a gallon per day in June
- **Monsoon advantage:** July–September ambient temps drop significantly — your hardest reservoir cooling period ends when peppers hit peak production

Growing Media for Hydroponic Peppers

Media	Best System	Notes	Local Source?
Perlite (70%) + Hydroton (30%)	Dutch Bucket ★	Best drainage and aeration. Reusable. Standard commercial pepper mix.	Purchase
Pure Perlite	Dutch Bucket, Ebb & Flow	Excellent drainage, very light. Needs more frequent irrigation cycles.	Purchase
Coarse granite grit (washed)	Dutch Bucket, DWC net pots	Free local alternative to hydroton. Rinse in dilute acid first to remove carbonates.	★ Local — crush granite
Washed local pumice	Any system	Volcanic pumice found in SE Arizona. Excellent drainage and aeration.	★ Local — Dragoon geology
Rockwool cubes (starter)	Seedling propagation	Best seedling medium for transitioning to hydroponics.	Purchase

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Starting Seeds & Transplanting

Indoor starts · heat mats · hardening off · timing

Peppers need a head start indoors — slow-growing and warmth-requiring. At Dragoon elevations, the last frost is April–May, so seeds started in February maximise your growing season. The key hydroponic difference: start in rockwool cubes rather than soil-based mix for a clean transition into your system.

Seed Starting Timeline by Variety

Variety Group	Start Indoors	Move to Hydro System	Reason
Rocoto / Manzano (C. pubescens)	January	Late April – May (4–5" tall, roots visible)	Slow germinators. Very long season (120+ days). Start earliest.
Habanero, Ghost (C. chinense)	Late January – Feb	Late April – May	Longest season of common varieties (85–100 days).
Jalapeño, Serrano, Cayenne (C. annuum)	Mid February	Late April – early May	Standard 8–10 week head start.
Poblano, Anaheim, Hatch, Bell	Mid February	Late April – early May	8–10 week head start.
Chiltepin (native)	February – March	May	Scarify seeds first (light sandpaper). Slow to establish.
Aji, Lemon Drop (C. baccatum)	February	Late April – May	8–10 week head start.

Germination & Propagation for Hydroponics

Factor	Target	Notes
Media	1.5" rockwool cubes ★	Soak cubes in pH 5.5 water before use. Rockwool transitions into hydro systems with no root disturbance.
Substrate temperature	80–85°F (C. annuum) 75–80°F (Rocoto)	Use a heat mat under seed trays. Most critical germination factor.
Germination time	7–21 days (C. annuum) 14–28 days (C. pubescens)	Do not give up early on Rocoto and habanero.
Initial nutrient solution	EC 0.4–0.6 — very dilute	Seedlings need almost no nutrients. Half-strength Masterblend or plain water.
pH of water/solution	5.8–6.0	Pre-treat rockwool with pH 5.5 water (acidic). Soak 1 hour before seeding.
Light after sprouting	16 hrs LED, 200–300 PPFD	Quantum board or T5 LED 2–4" above seedlings. Keep cool to prevent stretch.

Transitioning Seedlings into the Hydroponic System

1	Wait for visible roots — Move to system only when roots are visible at the base of the rockwool cube — usually when seedling is 3–5" tall.
2	Place cube in net pot — Set rockwool cube directly in a 3" net pot. Fill around with hydroton or perlite to support. No bare-rooting needed.
3	Drop reservoir level — For DWC: start with water level 1" below the net pot base. Roots will grow down to find the solution within days.
4	Start at low EC — First week in system: EC 0.8–1.0. Increase to vegetative level (1.4–1.8) once plant shows active new growth.
5	Protect from UV — Dragoon UV at elevation is intense. Provide 30% shade cloth for the first 2 weeks outdoors.

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Water Quality & Reservoir Management

Source water · bicarbonate pre-treatment · EC · reservoir changes

SE Arizona well water is among the hardest in the country — high in dissolved calcium, magnesium, and bicarbonates that constantly push pH upward. Pre-treating your water before mixing nutrients is the single most impactful practice for stable, predictable pepper production in the Dragoons.

Water Source Quality

Source	Baseline EC	Suitability	Recommendation
RO filtered water	~0.0	★★★★★ Perfect	Best baseline. Mix nutrients from zero. Most consistent results.
Monsoon rainwater (collected)	< 0.05	★★★★★ Excellent	Collect July–Sept. Store in dark sealed tanks. Nearly equal to RO.
RO / rain blend 50:50	0.1–0.2	★★★★ Very Good	Stretches RO supply while maintaining low baseline.
Treated well water	0.3–0.8	★★★ Usable	Requires full bicarbonate scrub. Reduce nutrient dose to account for baseline minerals.
Untreated well water	0.3–0.8	★★ Difficult	pH will be unstable all season. Invest in RO filter for best results.

Bicarbonate Pre-Treatment — Essential for Well Water

Bicarbonates in well water buffer pH upward constantly, fighting every correction you make. Neutralise them before adding nutrients — you will see fizzing as CO₂ releases. Wait until fizzing stops completely.

Step	Action	Target
1	Fill reservoir with source water	Note starting pH and EC
2	Add citric acid or phosphoric acid slowly while stirring	Fizzing = bicarbonates reacting and releasing CO ₂
3	Wait for fizzing to stop completely	All bicarbonates neutralised
4	Adjust pH down to 5.8–6.0	Now pH is stable — won't bounce back
5	Add Masterblend, then Epsom salt, then Calcium Nitrate	Always in this order, dissolved separately
6	Final EC and pH check	Target EC per stage table below

pH Targets for Hydroponic Peppers

- **Target pH: 5.8 – 6.2** — tighter than soil growing. Peppers tolerate up to 6.5 but micronutrient availability drops above 6.2
- **Blossom end rot in hydroponics** is almost always a pH or EC problem, not a calcium deficiency — check these first
- **Check pH daily** for the first 2 weeks of a new reservoir to learn your water's drift rate
- **pH Down (local options):** prickly pear vinegar for minor adjustments; citric acid stock solution for standard corrections; phosphoric acid for pre-treating hard well water

Reservoir Management Schedule

Task	Frequency	Notes
pH check	Daily	Adjust to 5.8–6.2. Note drift direction — tells you about plant uptake.
EC check	Daily	Top off with half-strength mix if EC drops (plants drinking more water than nutrients). Top off with plain water if EC rises (evaporation concentrating the solution).
Water level top-off	Daily	Never let roots dry out. High-desert evaporation is significant — even covered reservoirs lose water.
Full reservoir change	Every 7–10 days (fruiting) Every 14 days (veg)	Prevents salt buildup. Flush system with clean water between changes.
Root inspection	Weekly	Healthy = bright white. Brown/slimy = Pythium root rot — check reservoir temp first.
Meter calibration	Monthly	Calibrate pH meter with buffer solution (pH 4.0 and 7.0). Inaccurate readings are the #1 cause of unexplained crop problems.

Blossom Drop Prevention in Hydroponics

- **Reservoir stability is your best tool** — consistent EC and pH prevents the nutrient stress that triggers flower drop in soil growing
- **Temperature:** Flowers drop above 90°F daytime or below 55°F nights — the Dragoon monsoon resolves both; June pre-monsoon is highest risk
- **High EC at flowering:** EC above 3.0 during bloom causes osmotic stress and blossom drop — keep EC at 1.8–2.2 through flowering
- **Thrips damage flowers directly** — yellow sticky traps near flowering plants; inspect weekly
- **Monsoon advantage:** Cooler nights in July trigger a second major flush of flowering on plants that dropped buds in June heat

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Nutrient Solution Management

Masterblend programme · EC targets by stage · deficiency guide

Peppers are heavy feeders with evolving nutrient needs throughout their season. The Masterblend 3-part system (4-18-38 + Epsom salt + Calcium Nitrate) delivers everything peppers need. The key is adjusting concentration — not the ratio — as the plant moves from vegetative growth to fruiting. Peppers need more magnesium than tomatoes — double the Epsom salt dose once fruiting begins.

Masterblend Pepper Programme — Per Gallon

Growth Stage	Masterblend (g/gal)	Epsom Salt (g/gal)	Cal-Nitrate (g/gal)	Target EC (mS/cm)	Target pH
Seedling / cutting	1.2	0.6	1.2	0.8 – 1.0	5.8 – 6.0
Early vegetative	2.4	1.2	2.4	1.4 – 1.8	5.8 – 6.0
Late vegetative	2.8	1.4	2.8	1.8 – 2.2	5.8 – 6.2
First flower buds	2.8	1.4	2.8	1.8 – 2.2	5.8 – 6.2
Peak flowering	2.4	1.2	2.4	1.6 – 2.0	5.8 – 6.0
Fruit sizing	3.0	2.4 ★ increase Mg	3.0	2.0 – 2.6	6.0 – 6.2
Peak fruiting	3.0	2.4	3.0	2.2 – 2.8	6.0 – 6.2
Late harvest	2.0	1.0	2.0	1.4 – 2.0	6.0

Pepper Mg rule: Peppers are the heaviest magnesium users of all common hydroponic crops. Once fruit sizing begins, increase Epsom salt to 2.4 g/gal (double the standard Masterblend ratio). Magnesium deficiency — yellowing between green veins on older leaves — is the most common pepper deficiency in hydroponics.

Nutrient Deficiency Quick Reference

Symptom	Deficiency	Hydroponic Fix	pH Cause
Yellow between green veins, older leaves first	Magnesium (Mg)	Double Epsom salt to 2.4 g/gal for 2 weeks. Foliar spray 1 tbsp/gal.	Locked out below 5.5
Yellow between green veins, new growth first	Iron (Fe) or Mn	Check pH immediately — almost always pH > 6.5. Adjust to 5.8–6.0.	Locked out above 6.5
Blossom end rot (dark spot on fruit base)	Calcium (Ca)	Check pH is 5.8–6.2. Check EC not above 3.0. Ensure Cal-Nitrate was added last and dissolved fully.	Locked out below 5.5 or above 7.0

Purple/red leaf undersides	Phosphorus (P)	Check pH not below 5.0. Warm reservoir above 65°F. Usually resolves with pH correction.	Locked out below 5.0
Pale leaves, slow growth	Nitrogen (N) — low EC	Raise overall Masterblend dose by 0.4 g/gal. Check EC is above 1.4.	N/A — EC management issue
Dark green, thick leaves, stunted	Nitrogen excess — high EC	Dilute reservoir. Do partial water change. EC should not exceed 3.0 for peppers.	N/A — EC management issue
Curled, distorted new growth	Calcium (Ca) or Boron (B)	Check mixing order — Cal-Nitrate MUST go in last. If order was correct, try a very dilute boron foliar spray.	Locked out above 7.0

EC Drift Management

- **EC rising between top-offs:** Plants drinking more water than nutrients — top off with plain RO/rain water until EC returns to target
- **EC falling between top-offs:** Plants consuming nutrients heavily — top off with half-strength Masterblend mix
- **EC stable but pH rising daily:** Bicarbonate effect — pre-treat next reservoir fill with acid before adding nutrients
- **Maximum EC for peppers: 3.2 mS/cm** — above this, salt stress causes tip burn, blossom drop, and stunted fruit

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Seasonal Growing Calendar

Month-by-month tasks for 4,500 – 5,500 ft hydroponic growing

Month	Key Hydroponic Tasks	System Status
January	Start Rocoto and habanero in rockwool under LEDs. Set up heat mats. Mix dilute seedling solution (EC 0.6). Clean and test all meters.	Rocoto in rockwool; LED seedling stage
February	Start all <i>C. annuum</i> varieties in rockwool. Set up and test reservoirs. Order Masterblend, Epsom salt, and Calcium Nitrate. Order RO filter if needed.	Pepper seedlings under grow lights
March	Pot up seedlings as roots emerge from rockwool. Begin vegetative nutrient solution (EC 1.4–1.8). Inspect system for leaks. Set up monsoon water collection.	Seedlings 4–6" tall in net pots
April	Transition outdoor systems. Insulate/bury reservoirs before heat arrives. Install shade cloth. Watch for late frost — bring indoors if below 40°F.	Moving to outdoor systems
May	All plants in outdoor systems. EC at vegetative level (1.4–1.8). Daily pH and EC checks. Stake large varieties immediately.	Rapid vegetative growth
June	First flower buds — reduce EC slightly to 1.8–2.2. Reservoir management critical in pre-monsoon heat. Check reservoir temp daily — must stay below 72°F.	Flowering; blossom drop risk
July	MONSOON BEGINS. High humidity — watch for powdery mildew and aphid surges. Increase Epsom salt to 2.4 g/gal (fruit sizing). Rocoto flowering heavily — perfect conditions.	Fruit sizing; monsoon rain
August	Peak production — <i>C. annuum</i> harvest begins. Change reservoirs every 7 days during peak fruiting. EC at 2.2–2.8 for fruiting stage.	Harvest jalapeños, serranos, cayenne
September	★ BEST MONTH. Cool nights return — plant vigour surges. Harvest all fully ripe varieties. Rocoto at peak. Change reservoir weekly. Keep EC at fruiting level.	Peak harvest all varieties
October	Harvest continues. Frost watch begins. Move container systems indoors before first frost. Rocoto — pot up and bring inside. Begin transitioning to LED indoor winter grow if desired.	Final harvest; indoor transition
November	Overwinter selected plants indoors under LEDs. Reduce EC to 1.2–1.4 (semi-dormant maintenance). Dry and process harvest. Clean outdoor system for storage.	Indoor overwintering; preservation
December	Overwintered plants under LED at 16 hrs. EC 1.0–1.4 — minimal feeding. Plan next season. Order seeds. Review reservoir notes from past season.	Semi-dormant indoor growing

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Environmental Control & Canopy

VPD · temperature · LED lighting · staking · increasing capsaicin

Hydroponic peppers respond dramatically to environmental optimisation. The Dragoon Mountains give you naturally good VPD in fall, cool nights for fruit set, and high-altitude UV for flavour development — advantages no indoor-only grower can replicate. Combine these with good canopy management for exceptional yields.

VPD Targets for Hydroponic Peppers

Stage	Temp	Humidity	VPD (kPa)	Dragoon Season
Seedling	72–77°F	70–80%	0.4–0.8	Indoor Jan–Mar
Early vegetative	72–82°F	60–70%	0.8–1.1	Indoor/outdoor Apr–May
Late veg / pre-flower	75–85°F	55–65%	1.0–1.4	May–June
Flowering	70–82°F	50–60%	1.2–1.6	June–July (monsoon helps)
Peak fruiting	75–85°F	50–60%	1.2–1.6	July–Sept (ideal naturally)
Late harvest	65–78°F	45–55%	1.4–1.8	Sept–Oct (Dragoon fall peak)

LED Lighting for Indoor & Greenhouse Pepper Growing

Scenario	PPFD Target	DLI	Photoperiod	Notes
Indoor seedlings	150–250 $\mu\text{mol}/\text{m}^2/\text{s}$	8–14	16–18 hrs	T5 or small quantum board 4" above canopy.
Indoor vegetative	400–600 $\mu\text{mol}/\text{m}^2/\text{s}$	20–30	16–18 hrs	Spider Farmer SF-2000 or equivalent. 18–24" above canopy.
Indoor fruiting	600–800 $\mu\text{mol}/\text{m}^2/\text{s}$	25–35	14–16 hrs	High-output quantum board. Drop to 16 hrs to simulate seasonal change.
Greenhouse supplement	200–400 $\mu\text{mol}/\text{m}^2/\text{s}$ (supplemental)	Adds 8–15 to natural DLI	2–4 hrs extension only	LED bar lights at dawn/dusk to extend photoperiod. Very cost effective.
Rocoto (summer)	200–400 $\mu\text{mol}/\text{m}^2/\text{s}$ (filtered)	12–18	14–16 hrs	Rocoto needs shade in summer. 30–40% shade cloth + LED in morning/evening gives ideal levels.

Canopy Management for Hydroponic Peppers

- **Stake at transplant to net pot** — use a bamboo cane in the bucket/net pot alongside the plant from day one
- **Remove first flower buds for 2–3 weeks** — redirects energy to root development in the hydroponic media, significantly improving total season yield
- **Pinch growing tip at 6–8" height** — forces branching and doubles fruiting sites
- **Remove leaves below the first fork** — improves airflow around reservoir opening, reducing algae risk
- **Rocoto training:** train the main stem along a trellis — these become large semi-vining plants in hydroponic systems and can produce hundreds of fruits in year 2+

How to Maximise Capsaicin in Hydroponics

- **Controlled water stress, final 2 weeks:** reduce reservoir topping-off frequency slightly. Mild stress — roots still wet — significantly increases capsaicin without damaging the plant
- **Reduce nitrogen after fruit set:** drop Masterblend from 3.0 to 2.0 g/gal in the last 3 weeks while maintaining Epsom salt and Cal-Nitrate at full rate
- **High PPFD accelerates ripening and capsaicin concentration:** increase to 700–800 $\mu\text{mol}/\text{m}^2/\text{s}$ in final 2–3 weeks
- **Let fruit ripen fully:** red jalapeños have 2–3× the capsaicin of green ones. The extra 3–4 weeks is worth it
- **Grow native Chiltepin:** the most intense local variety, thrives in DWC, and produces prolifically with minimal inputs

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Pest & Disease Management

Common threats · organic controls · Dragoon-specific issues

Pest / Problem	Season	Signs	Organic Control
Aphids ★ Most common	Spring & Fall	Dense clusters on new growth; sticky honeydew; ants farming them	Strong water blast. Insecticidal soap spray. Neem oil. Plant basil nearby. Ladybugs (buy or attract naturally).
Spider mites	June (hot dry)	Tiny webbing under leaves; bronzed, stippled foliage	Increase humidity around plants (mist in morning). Neem oil or insecticidal soap. Avoid broad-spectrum pesticides.
Thrips	Spring–Summer	Silvery scars on leaves and fruit; distorted new growth; dropped flowers	Yellow sticky traps. Spinosad spray. Remove affected leaves. Blue sticky traps specifically target thrips.
Pepper weevil	Summer	Small holes in green fruit; fruit drops prematurely; internal larvae	Remove dropped fruit immediately. Neem oil spray on developing fruit. Pyrethrin as last resort.
Cutworms	Spring — transplant time	Seedlings cut off at soil level overnight	Cardboard collar around each stem 2" into soil. BT (<i>Bacillus thuringiensis</i>) drench at base. Diatomaceous earth.
Grasshoppers	June–Sept	Rapid defoliation — especially severe in drought years	<i>Nosema locustae</i> biological bait. Row covers. Chickens if nearby. Kaolin clay spray on leaves.
Bacterial wilt (Phytophthora)	Post-monsoon rain	Sudden total wilting despite wet soil; dark basal crown	No cure — remove plant and soil immediately. Prevent: don't overwater; improve drainage; rotate solanaceae every 3 years.
Blossom end rot	Peak fruiting	Dark sunken spot on fruit bottom; watery, then dry and brown	See Chapter 7 — consistent watering + pH 6.0–7.0 prevents nearly all cases.
Sun-scald	June–August	White papery patches on fruit facing the sun	30% shade cloth in peak heat. Leave enough foliage to shade fruit. Orient rows north-south so plants shade each other.
Powdery mildew	Monsoon season	White powder on leaves — mainly in dense canopy	Improve airflow around plants. Dilute milk spray (1 part milk : 9 parts water). Potassium bicarbonate spray. Increase plant spacing.
Javelina	Year-round	Entire plants uprooted or eaten; strong musk smell	Low-voltage electric fence around bed perimeter. Welded wire cage around individual plants in first season.

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Harvest, Drying & Processing

When to pick · green vs. red · drying · powder · sauce

Most peppers can be harvested at two very different stages — green (unripe) or fully coloured (ripe) — and each gives a completely different flavour, heat level, and culinary use. The Dragoon fall season gives you excellent natural drying conditions for making chile powder and ristras.

Harvest Guide by Variety

Variety	Harvest Green?	Harvest Ripe?	Colour at Ripe	Dragoon Timing
Jalapeño	Yes — firm, dark green	Yes — red, sweeter and hotter	Red	Aug–Oct
Poblano	Yes ★ — classic use; dark green = poblano	Yes — dried red = ancho chile	Red/Dark Red	Aug–Oct
Serrano	Yes — green has brighter flavour	Yes — red/orange is hotter	Red or Orange	Aug–Oct
Cayenne	No — wait for red	Yes ★ — dry when fully red	Red	Sept–Oct
Anaheim / Hatch	Yes ★ — roast green	Yes — dried red = California chile	Red	Aug–Sept
Habanero	No — wait for full colour	Yes ★	Orange/Red/C hocolate	Sept–Oct
Chiltepin	No — wait for red	Yes ★	Red	Sept–Oct
Rocoto	No — wait for full colour	Yes ★	Red/Orange/Y ellow	Sept–Oct (after monsoon cools)
Bell Pepper	Yes — green bell	Yes — red/yellow/orange; sweeter	Red/Yellow/Or ange	Aug–Oct

Drying Methods for the Dragoon Climate

The Dragoon fall season — dry, warm days with low humidity — is perfect for traditional sun-drying and ristra-hanging. This is exactly the climate New Mexico chile growers prize.

Method	Best Varieties	How To	Time
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Sun drying ★ Best in Dragoons	Cayenne, Chiltepin, Serrano, thin-walled chiles	Lay on wire racks in full sun. Bring in at night. Turn daily. Done when fully dry and brittle.	7–14 days
Ristra (hanging)	Hatch, Anaheim, Cayenne, Chiltepin	Thread through stems with needle and twine. Hang in shade with good airflow. Traditional Southwest method.	3–6 weeks
Oven drying	All varieties	Slice or leave whole. 150–170°F with door cracked. Check every 2 hours.	4–12 hours
Dehydrator	All varieties	125–135°F setting. Slice thick-walled varieties. Single layer on trays.	8–24 hours
Smoke drying (chipotle)	Jalapeño → Chipotle	Cold-smoke over local mesquite or oak 2–3 days. Dragoon area has excellent mesquite for this.	2–3 days

★ **Local tip:** Local mesquite is abundant in the lower Dragoon foothills and Sulphur Springs Valley. Slow-smoking red jalapeños over mesquite wood for 2–3 days produces traditional chipotle with a distinct Dragoon terroir — an artisan product worth making.

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Overwintering Peppers

Indoor storage · containers · perennial management

Pepper plants are perennial in frost-free climates. At Dragoon elevation, frost ends outdoor growing in October–November, but bringing selected plants indoors for winter restarts them 6–8 weeks ahead of a new plant the following spring — and Rocoto plants are worth keeping alive for decades.

Which Varieties Are Worth Overwintering?

Variety	Worth Overwintering?	Why
Rocoto / Manzano (C. pubescens)	★★ Absolutely — treat as a permanent plant	Lives 15+ years. Gets more productive each year. Too slow-growing to replace annually.
Habanero / Scotch Bonnet	★★ Yes — strongly recommended	Takes 2+ years to reach peak production. Year-2 plants produce 3–4× more than seedlings.
Ghost Pepper / Extreme heat	★★ Yes — the only way to get reliable harvest	Needs 100–120 day season. Year-2 plants already at full size at bud break.
Chiltepin (native)	★ Yes, or let self-seed in protected microclimate	Perennial in Dragoons if sited against a south-facing wall or rock. Some plants survive mild winters in protected spots.
Jalapeño, Serrano, Cayenne	Optional — mainly for fun	Easy to grow from seed. Overwintering gives 4–6 week head start. Low priority.
Bell Pepper	Not worth it	Inexpensive seedlings. Poor indoor performer. Start fresh annually.

Overwintering Protocol for Hydroponic Plants

1	Cut back hard before bringing in — In early October, prune plants back to 6–8" stubs, leaving 3–4 main branches. This reduces transpiration and makes indoor plants manageable.
2	Inspect and treat for pests — Check carefully for mites, aphids, and scale before bringing inside. Treat with insecticidal soap spray and let dry before moving indoors.
3	Keep in hydroponic container — Leave Rocoto and habanero in their Dutch buckets with perlite media. Prune back hard to 8–10" stubs. Much easier than repotting.
4	Reduce nutrient solution to maintenance — EC 1.0–1.2 with standard Masterblend ratio. Change reservoir every 3–4 weeks over winter — plants are semi-dormant and uptake is minimal.
5	Water sparingly — Every 7–10 days — just enough to prevent complete desiccation. No fertiliser until February.
6	Resume feeding in February — Start fish emulsion or compost tea in February. Plants will push new growth rapidly.

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Harden off and return outdoors in May — Acclimatise for 1–2 weeks as you would a seedling. Year-2 plants will explode into production within weeks of going back out.

The Dragoon Mountains are ancestral pepper country. The wild Chiltepin has grown here for millennia, and the cool nights, monsoon rains, intense fall sunshine, and mineral-rich soils create conditions that produce peppers of exceptional flavour, heat, and character. From native Chiltepin growing wild in the washes to Andean Rocoto thriving in the cool evening air, this landscape was made for chile.